

D772

Statistical Data Literacy

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A school surveyed 240 families about their preferred way to receive school news: email, text, paper flyer, or app. The survey was sent in October and took five minutes. Which display fits the data best?

- A. A bar graph of the number of families choosing each method.
- B. A line graph of survey minutes by month.
- C. A line graph connecting email, text, flyer, and app.
- D. A histogram of the methods.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Five employees at a small company earn \$40,000, \$42,000, \$45,000, \$48,000, and \$250,000. The company is ten years old and has an office downtown. Which value best describes a typical salary?

- A. The mode, since every value is typical.
- B. The mean, \$85,000.
- C. The range, \$210,000.
- D. The median, \$45,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A histogram of home prices has a long tail to the right. How do the mean and median most likely compare?

- A. The mean is greater than the median.
- B. They are always equal.
- C. The median is greater than the mean.
- D. The mode is greater than both.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Two classes took the same test. Both had a mean of 78. Class A's standard deviation was 4; Class B's was 12. Class B met in a newer building with larger windows. What can you conclude?

- A. Class A had more variation.
- B. The newer building raised Class B's scores.
- C. Class B scored higher on average.
- D. Class B's scores were more spread out from the mean than Class A's.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A news website asks readers to vote online on whether the town should build a skate park. It receives 5,000 votes, 80% against. The site has a modern design and loads quickly on phones. What is the main problem with using this result to describe all town residents?

- A. The site design caused the result.
- B. There is no problem, because 80% is a clear majority.
- C. It is a voluntary-response sample, so people with strong opinions are overrepresented.
- D. 5,000 votes is too small a sample.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

An observational study finds that students who eat breakfast at school have higher test scores. The study included 1,200 students and was published in a popular magazine. Which conclusion is justified?

- A. Breakfast has no effect on scores.
- B. Breakfast and scores are associated, but other factors could explain both.
- C. Eating breakfast causes higher scores.
- D. The result proves cause because 1,200 is a large sample.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A researcher wants to show that a new tutoring program causes higher math scores. Which design supports that claim?

- A.** Measure only the tutored students before and after.
- B.** Compare scores of students who chose tutoring with those who did not.
- C.** Randomly assign students to tutoring or no tutoring and compare scores.
- D.** Survey tutors about how much students improved.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A bar graph shows School A's attendance rate at 94% and School B's at 96%. The vertical axis starts at 93%, so B's bar looks three times taller. The graph uses the schools' colors. What is the accurate reading?

- A.** The colors make the graph misleading.
- B.** B's attendance is three times A's.
- C.** B's rate is 2 percentage points higher; the truncated axis exaggerates the gap.
- D.** B's rate is 3 points higher.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Scores on a reading test are roughly normal with a mean of 70 and a standard deviation of 8. The test was given to 600 students in a gym. About what percent of students scored between 54 and 86?

- A.** About 68%
- B.** About 99.7%
- C.** About 570 students
- D.** About 95%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Using the same test (mean 70, SD 8, roughly normal), about what percent of students scored above 86?

- A. About 5%
- B. About 2.5%
- C. About 95%
- D. About 16%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

For a set of commute times, $Q1 = 20$ minutes and $Q3 = 32$ minutes. One worker, who drives a hybrid and lives near a lake, reports 52 minutes. Is 52 an outlier by the $1.5 \times IQR$ rule?

- A. Yes; it is above the upper fence of 50 minutes.
- B. No; hybrid drivers commute longer distances.
- C. No; the upper fence is 56 minutes.
- D. No; the fence is $1.5 \times IQR$ above the median, not $Q3$.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

With $Q1 = 20$ and $Q3 = 32$, below what value would a data point be a low outlier?

- A. 14
- B. 2
- C. 8
- D. 38

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A school surveyed 240 families about their preferred way to receive school news: email, text, paper flyer, or app. ~~The survey was sent in October and took five minutes.~~ Which display fits the data best?

Target word: graph for categorical data **Decides:** Preferred method is a category.

A	Correct	Bar graphs compare counts across categories.
B	Camouflage	Graphs the irrelevant timing detail.
C	Adjacent category	Line graphs show change over time, not unordered categories.
D	Confusion set	Histograms need quantitative data with numeric bins.

2. Answer D Concept 2

Five employees at a small company earn \$40,000, \$42,000, \$45,000, \$48,000, and \$250,000. ~~The company is ten years old and has an office downtown.~~ Which value best describes a typical salary?

Target word: choose center with an outlier **Decides:** One extreme value; median resists it.

A	Absolute language	No value repeats, so there is no mode.
B	Dropped element	Correct mean ($\$425,000 / 5$), but it ignores the effect of the outlier.
C	Adjacent category	Range measures spread, not center.
D	Correct	The median is not pulled by the \$250,000 salary.

3. Answer A Concept 2 · transfer

A histogram of home prices has a long tail to the right. How do the mean and median most likely compare?

Target word: direction of skew effect **Decides:** Mean is pulled toward the right tail.

A	Correct	A few high values pull the mean up.
B	Absolute language	Only in symmetric distributions are they about equal.
C	Inverted operation	That happens with a left tail.
D	Confusion set	In right skew the mode is usually lowest.

4. Answer D Concept 3

Two classes took the same test. Both had a mean of 78. Class A's standard deviation was 4; Class B's was 12. ~~Class B met in a newer building with larger windows.~~ What can you conclude?

Target word: interpret SD **Decides:** Same mean; B has more spread.

A	Inverted operation	A smaller SD means less spread.
B	Camouflage	The building does not decide the comparison.
C	Confusion set	The means are equal; SD is not the average.
D	Correct	A larger SD means more variation.

5. Answer C Concept 4

A news website asks readers to vote online on whether the town should build a skate park. It receives 5,000 votes, 80% against. ~~The site has a modern design and loads quickly on phones.~~ What is the main problem with using this result to describe all town residents?

Target word: sampling bias **Decides:** Voluntary response sample.

A	Camouflage	Design is irrelevant to who chose to respond.
B	Absolute language	A lopsided result in a biased sample proves nothing about the town.
C	Correct	Self-selected samples rarely represent the population.
D	Wrong stem number	Size is not the issue; a large biased sample is still biased.

6. Answer B Concept 5

An observational study finds that students who eat breakfast at school have higher test scores. ~~The study included 1,200 students and was published in a popular magazine.~~ Which conclusion is justified?

Target word: what an observational study shows **Decides:** No random assignment, so only an association.

A	Absolute language	The study cannot rule out an effect either.
B	Correct	Observational data show association; lurking variables are possible.
C	Inverted operation	Claims cause without an experiment.
D	Camouflage	Sample size does not turn association into cause.

7. Answer C Concept 5 • transfer

A researcher wants to show that a new tutoring program causes higher math scores. Which design supports that claim?

Target word: design that supports cause **Decides:** Random assignment to treatment and control.

A	Confusion set	No control group; other changes could explain gains.
B	Dropped element	Self-selection leaves lurking variables.
C	Correct	Random assignment balances other factors, supporting a causal claim.
D	Adjacent category	Opinions are not a controlled comparison.

8. Answer C Concept 6

A bar graph shows School A's attendance rate at 94% and School B's at 96%. The vertical axis starts at 93%, so B's bar looks three times taller. ~~The graph uses the schools' colors.~~ What is the accurate reading?

Target word: read past a truncated axis **Decides:** The actual difference is 2 points.

A	Camouflage	Color is not the distortion; the axis is.
B	Inverted operation	Takes the distorted bar height at face value.
C	Correct	Read the numbers, not the bar heights.
D	Wrong stem number	$96 - 94 = 2$.

9. Answer D Concept 7

Scores on a reading test are roughly normal with a mean of 70 and a standard deviation of 8. ~~The test was given to 600 students in a gym.~~ About what percent of students scored between 54 and 86?

Target word: empirical rule percent **Decides:** 54 and 86 are 2 SD from the mean.

A	Confusion set	That is within 1 SD (62 to 78).
B	Wrong stem number	That is within 3 SD (46 to 94).
C	Camouflage	Uses the 600 detail; the question asks for a percent.
D	Correct	70 ± 16 is within 2 SD.

10. Answer B Concept 7 · transfer

Using the same test (mean 70, SD 8, roughly normal), about what percent of students scored above 86?

Target word: one tail beyond 2 SD **Decides:** 5% outside ± 2 SD, split into two tails.

A	Dropped element	Forgets to split the 5% between two tails.
B	Correct	Half of the 5% outside 2 SD is in the upper tail.
C	Inverted operation	That is the percent inside the range.
D	Confusion set	That is the tail beyond 1 SD.

11. Answer A Concept 8

For a set of commute times, $Q1 = 20$ minutes and $Q3 = 32$ minutes. One worker, ~~who drives a hybrid and lives near a lake,~~ reports 52 minutes. Is 52 an outlier by the $1.5 \times IQR$ rule?

Target word: apply the outlier rule **Decides:** Upper fence = $32 + 1.5 \times 12 = 50$.

A	Correct	$IQR = 12$; $1.5 \times 12 = 18$; $32 + 18 = 50 < 52$.
B	Camouflage	The car type is irrelevant.
C	Wrong stem number	Adds $2 \times IQR$ (24) instead of $1.5 \times IQR$.
D	Confusion set	The rule measures from $Q3$, not the median.

12. Answer B Concept 8 · transfer

With $Q1 = 20$ and $Q3 = 32$, below what value would a data point be a low outlier?

Target word: lower fence **Decides:** $20 - 1.5 \times 12 = 2$.

A	Confusion set	Subtracts 18 from $Q3$ instead of $Q1$.
B	Correct	$Q1 - 18 = 2$.
C	Dropped element	Subtracts IQR (12) without multiplying by 1.5.
D	Inverted operation	Adds 18 to $Q1$ instead of subtracting.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Variable type decides the graph	Categorical data (groups, labels) go in bar or pie charts; quantitative data (measured numbers) go in histograms, dot plots, or box plots; change over time goes in a line graph.	Using a histogram for categories or a line graph for unrelated groups.	“Favorite,” “type,” or “group” vs. a measured amount.
2	Center in skewed data	The mean is pulled toward a long tail or extreme value; the median resists it, so use the median to describe the typical value in skewed data.	Reporting the mean as “typical” when a few extreme values exist.	One or a few values far from the rest.
3	What standard deviation shows	Standard deviation measures typical distance from the mean; larger SD means more spread, not a higher average.	Reading a larger SD as better or higher performance.	Two groups with the same mean and different SDs.
4	Sampling bias	Conclusions generalize only when the sample represents the population; voluntary-response and convenience samples often overrepresent strong opinions or easy-to-reach people.	Trusting a large sample even when it is self-selected.	An online poll, a call-in survey, or surveying whoever is nearby.
5	Correlation vs. causation	An observed association does not show cause; only a randomized experiment can support a causal claim, because lurking variables may drive both.	Concluding cause from an observational study.	“Students who ... scored higher,” with no random assignment.
6	Misleading graphs	A vertical axis that does not start at zero, uneven scales, or 3D effects can make small differences look large; read the actual numbers.	Judging change by bar height alone.	An axis that starts at a number well above zero.
7	Empirical rule	In a roughly normal distribution, about 68% of data fall within 1 SD of the mean, 95% within 2 SD, and 99.7% within 3 SD.	Using 68% for two standard deviations, or forgetting to split the tails in half.	A mean, an SD, and a bell-shaped distribution.
8	IQR and the outlier rule	$IQR = Q3 - Q1$; a value is an outlier if it is more than $1.5 \times IQR$ below $Q1$ or above $Q3$.	Adding $1.5 \times IQR$ to the median, or using the range.	Quartiles and a question about unusual values.

Course scope (public description): Statistical Data Literacy utilizes concepts in statistics, research, data, and data representation to explore the interpretation of data and their application to decision-making. The course uses real-world scenarios to build upon foundational principles of data collection and description. The use of data to make and explain decisions is also explored. The course also builds on skills of descriptive statistics and graphical depictions of data to help students understand data and make data-based decisions. This course has no prerequisites.